INCH-POUND

MS16629E
14 March 2012
SUPERSEDING
MS16629D
20 April 1990

DETAIL SPECIFICATION SHEET

RING, RETAINING, INTERNAL, BOWED
(TAPERED SECTION TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

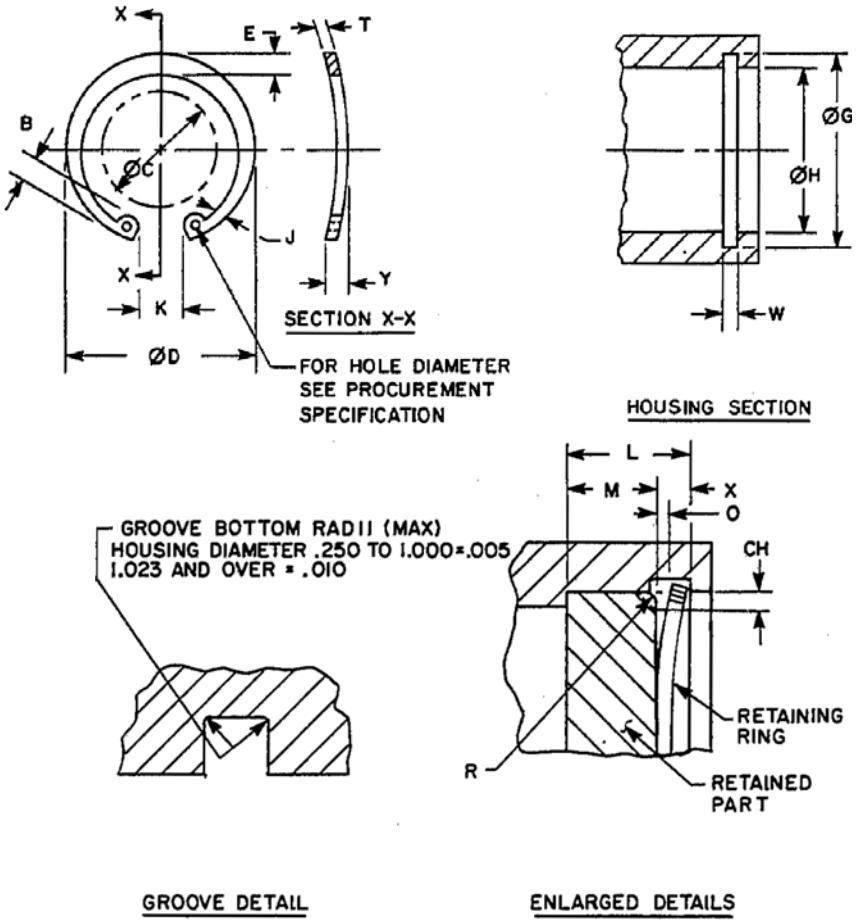


FIGURE 1. RING, RETAINING.

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TABLE I DIMENSIONS.

ØH HOUSING (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		Y OVERALL BOW HEIGHT		ØG RECOMMENDED GROOVE (REF)		W WIDTH		K 3/	ØC 4/ CLEAR	X		O 5/ TAKE UP OF L AND M	R 6/ CH 6/ OF RETAINED PART (REF)	
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MIN		MAX	MIN	MAX	MAX	MAX
.250	6.4	.280		.065		.025	±.002	.015	±.002	.015		.036	±.006	.268	±.001	.030	+.002	.047	.115	.020	.028		.011	.0085
.312	7.7	.346		.066		.033		.018		.015				.330	.0015	.030	-.000	.055	.173	.020	.028		.016	.013
															FIM 2/									
.375	9.5	.415		.082	±.003	.040		.028		.025				.397	±.002	.040		.063	.204	.030	.038	.008	.023	.018
.438	11.1	.482		.098		.049	±.003	.029	±.003	.025		.047	±.006	.461	.002	.040		.063	.23	.030	.038		.027	.021
.453	11.5	.498		.098		.050		.030		.025				.477	FIM 2/	.040		.071	.25	.030	.038		.027	.021
.500	12.7	.548	±.010	.114		.053		.035		.035				.530		.055		.090	.26	.042	.053		.027	.021
.512	13.0	.560		.114		.053		.035		.035				.542		.055		.092	.27	.042	.053		.027	.021
.562	14.3	.620		.132		.053	±.004	.035	±.004	.035		.063	±.007	.596	±.002	.055		.095	.275	.042	.053		.027	.021
.625	15.9	.694		.132		.060		.035		.035				.665	.004	.055		.104	.34	.042	.053		.027	.021
.688	17.5	.763		.132		.063		.036		.035				.732	FIM 2/	.055		.118	.40	.042	.053		.027	.021
.750	19.0	.831		.142		.070		.040		.035				.796		.055		,143	.45	.042	.053		.032	.025
.777	19.7	.859		.146		.074		.044		.042				.825		.062		.145	.475	.049	.060		.035	.028
.812	20.6	.901		.155		.077		.044		.042				.862		.062		.153	.49	.049	.060		.035	.028
.866	22.0	.961		.155		.081		.045		.042				.920		.062		.172	.54	.049	.060		.035	.028
.875	22.2	.971		.155		.084	±.005	.045	±.005	.042		.073	±.008	.931	±.003	.062		.179	.545	.049	.060		.035	.028
															.004									
.901	22.9	1.000	±.015	.155		.087		.047		.042	±.002			.959	FIM 2/	.062		.188	.565	.049	.060		.038	.030
.938	23.6	1.041		.155		.091		.050		.042				1.000		.062	+.003	.200	.61	.049	.060		.038	.030
1.000	25.4	1.111		.155		.104		.052		.042				1.066		.062	-.000	.212	.665	.049	.060		.042	.034
1.023	26.0	1.136		.155	±.005	.106		.054		.042				1.091		.062		.220	.69	.049	.060	.011	.042	.034
1.062	27.0	1.180		.180		.110		.055		.050				1.130		.070		.213	.685	.057	.068		.044	.035
1.125	28.6	1.249		.180		.116		.057		.050				1.197		.070		.232	.745	.057	.068		.047	.036
1.181	30.0	1.319		.180		.120		.058		.050				1.255		.070		.226	.79	.057	.068		.047	.036
1.188	30.2	1.319		.180		.120		.058		.050				1.262		.070		.245	.80	.057	.068		.047	.036
1.250	31.7	1.388		.180		.124		.062		.050				1.330	±.004	.070		.265	.875	.057	.068		.048	.038
1.259	32.0	1.388	±.025	.180		.124	±.006	.062	±.006	.050		.085	±.012	1.339	.005	.070		.290	.885	.057	.068		.048	.038
1.312	33.3	1.456		.180		.124		.062		.050				1.396	FIM 2/	.070		.284	.93	.057	.068		.048	.038
1.375	34.9	1.526		.180		.130		.063		.050				1.461		.070		.297	.99	.057	.068		.048	.038
1.378	35.0	1.526		.180		.130		.063		.050				1.464		.070		.305	.99	.057	.068		.048	.038
1.438	36.5	1.596		.180		.133		.065		.050				1.528		.070		.313	1.06	.057	.068		.048	.038
1.456	37.0	1.616		.180		.133		.065		.050				1.548		.070		.320	1.08	.057	.068		.048	.038
1.500	38.1	1.660		.180		,133		.066		.050				1.594		.070		.340	1.12	.057	.068		.048	.038

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE HOUSING.

3/ K - MNIMUM GAP WIDTH WHEN THE RING IS PROPERLY SEATED IN THE GROOVE. (DESIGN REFERENCE DIMENSION).

4/ C - ACTUAL CLEARANCE DIAMETER WHEN THE RING IS SPRUNG INTO THE HOUSING, PRIOR TO INSTALLATION INTO THE GROOVE. (DESIGN REFERENCE DIMENSION).

5/ O - END PLAY TAKE UP (DESIGN REFERENCE DIMENSION).

6/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RING. THRUST LOADS OF RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS ARE TABULATED ON PAGE 6.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE II, CLASS 1 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

   (a) CARBON SPRING STEEL,GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568/A568M OR ASTM A684/A684M.
   (b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700).
   (c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS

TABLE II. HARDNESS.

ØH HOUSING (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.250 TO .750 INCL	85.5-87.9 HR15N 1/ (50-55 HRC EQUIV)	83.9-85.9 HR15N 1/ (46-51 HRC EQUIV)	79.9-82.0 HR15N 1/ (39-43 HRC EQUIV)
.777 TO .901 INCL	66.7-71.2 HR30N 1/ (48-53 HRC EQUIV)	64.8-69.5 HR30N 1/ (46-51 HRC EQUIV)	58.6-62.2 HR30N 1/ (39-43 HRC EQUIV)
938 TO 1.500 INCL	52.5-58.6 HR45N 1/ (48-53 HRC EQUIV)	50.3-56.1 HR45N 1/ (46-51 HRC EQUIV)	41.9-46.7 HR45N 1/ (39-43 HRC EQUIV)

1/ USE HUGHEST SCALE TO PROVIDE SECTION WIDTH EQUAL TO OR GREATER THAN 5 TIMES THE BRAILE IMPRESSION DIAMETER.

4. PROTECTIVE FINISH OR TREATMENT:

   (a) CARBON STEEL – SHALL BE AS SPECIFIED (SEE TABLE III OR IV):

      (1) CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5.

      (2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 OR ASTM B695, TYPE II, CLASS 5.

      (3) PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

   (b) CORROSION RESISTANT STEEL – SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III AND IV.

   EXAMPLE: MS16629-1100 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATED, INTERNAL, BOWED RETAINING RING FOR USE IN A 1.000 DIAMETER HOUSING.

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TABLE III. DASH NUMBERS FOR MS16629.

ØH HOUSING (REF)	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL CORROSION RESISTANT	1/ BERYLLIUM COPPER
	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO
.250	-1025	-2025	-3025	-4025	-5025
.312	-1031	-2031	-3031	-4031	-5031
.375	-1037	-2037	-3037	-4037	-5037
.438	-1043	-2043	-3043	-4043	-5043
.453	-1045	-2045	-3045	-4045	-5045
.500	-1050	-2050	-3050	-4050	-5050
.512	-1051	-2051	-3051	-4051	-5051
.562	-1056	-2056	-3056	-4056	-5056
.625	-1062	-2062	-3062	-4062	-5062
.688	-1068	-2068	-3068	-4068	-5068
.750	-1075	-2075	-3075	-4075	-5075
.777	-1077	-2077	-3077	-4077	-5077
.812	-1081	-2081	-3081	-4081	-5081
.866	-1086	-2086	-3086	-4086	-5086
.875	-1087	-2087	-3087	-4087	-5087
.901	-1090	-2090	-3090	-4090	-5090
.938	-1093	-2093	-3093	-4093	-5093
1.000	-1100	-2100	-3100	-4100	-5100
1.023	-1102	-2102	-3102	-4102	-5102
1.062	-1106	-2106	-3106	-4106	-5106
1.125	-1112	-2112	-3112	-4112	-5112
1.181	-1118 2/	-2118 2/	-3118 2/	-4118 2/	-5118 2/
1.188	-1118 2/	-2118 2/	-3118 2/	-4118 2/	-5118 2/
1.250	-1125 2/	-2125 2/	-3125 2/	-4125 2/	-5125 2/
1.259	-1125 2/	-2125 2/	-3125 2/	-4125 2/	-5125 2/
1.312	-1131	-2131	-3131	-4131	-5131
1.375	-1137 2/	-2137 2/	-3137 2/	-4137 2/	-5137 2/
1.378	-1137 2/	-2137 2/	-3137 2/	-4137 2/	-5137 2/
1.438	-1143	-2143	-3143	-4143	-5143
1.456	-1145	-2145	-3145	-4145	-5145
1.500	-1150	-2150	-3150	-4150	-5150

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

2/ SAME DASH NUMBERS SUITABLE FOR EITHER HOUSING DIAMETER (INCH OR MM).

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TABLE IV. SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS.

	INACTIVE	SUBSTITUTE	SUBSTITUTE	SUBSTITUTE
ØH HOUSING (REF)	CARBON STEEL	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT
	MS16629	MS16629	MS16629	MS16629
.250	-25	-1025	-2025	-3025
.312	-31	-1031	-2031	-3031
.375	-37	-1037	-2037	-3037
.438	-43	-1043	-2043	-3043
.453	-45	-1045	-2045	-3045
.500	-50	-1050	-2050	-3050
.512	-51	-1051	-2051	-3051
.562	-56	-1056	-2056	-3056
.625	-62	-1062	-2062	-3062
.688	-68	-1068	-2068	-3068
.750	-75	-1075	-2075	-3075
.777	-77	-1077	-2077	-3077
.812	-81	-1081	-2081	-3081
.866	-86	-1086	-2086	-3086
.875	-87	-1087	-2087	-3087
.901	-90	-1090	-2090	-3090
.938	-93	-1093	-2093	-3093
1.000	-100	-1100	-2100	-3100
1.023	-102	-1102	-2102	-3102
1.062	-106	-1106	-2106	-3106
1.125	-112	-1112	-2112	-3112
1.181	-118	-1118	-2118	-3118
1.188	-118	-1118	-2118	-3118
1.250	-125	-1125	-2125	-3125
1.259	-125	-1125	-2125	-3125
1.312	-131	-1131	-2131	-3131
1.375	-137	-1137	-2137	-3137
1.378	-137	-1137	-2137	-3137
1.438	-143	-1143	-2143	-3143
1.456	-145	-1145	-2145	-3145
1.500	-150	-1150	-2150	-3150

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE:

(a) INTENDED USE- TO PROVIDE UNIFORM PROTRUDING SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS IN HOUSING (BORES). THE RINGS ARE BENT LIKE A BOW OUT OF PLANE. FREE ENDS AND OPPOSITE EDGE ABUT OUTER GROOVE WALL. MIDSECTION OF RING ABUTS MACHINE PART, RING WILL TAKE UP END PLAY RESILIENTLY. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE – THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING: RINGS SHOULD NOT BE EXCESSIVELY CONTRACTED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE OUTSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN EXCESSIVELY CONTRACTED, (PROVIDING THE GROOVE DIAMETER HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. (ABUTTING COMPONENTS TO HAVE SHARP CORNERS)=

$$P = \frac{\pi HTX}{F}$$

WHERE:

P = ALLOWABLE THRUST LOAD (POUNDS)
H = HOUSING DIAMETER (INCHES)
T = RING, THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A FACTOR OF SAFETY, F=4, IS RECOMMENDED, SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES, BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL =

$$P = \frac{\pi HdY}{F}$$

WHERE:

P = ALLOWABLE COMPRESSION LOAD (POUNDS)
H = HOUSING DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD A SAFETY FACTOR, F = 2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF HOUSING =

$$Z = 3d$$

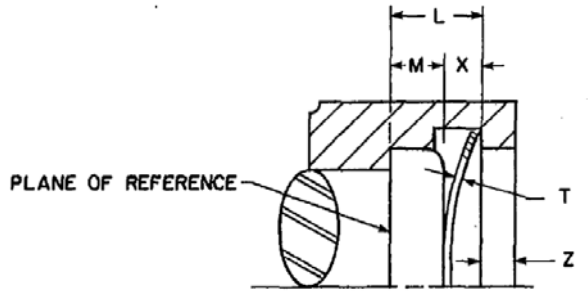
WHERE:

Z = MINIMUM DISTANCE BEWTWEEN OUTER GROOVE WALL AND END OF HOUSING (INCHES)
d = GROOVE DEPTH (INCHES)

(e) LOCATION OF GROOVE =

$$L_{MAX} = M_{MIN} + X_{MAX}$$

$$L_{MIN} = M_{MAX} + X_{MIN}$$



1/ X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .750 INCH HOUSING DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .777 INCH AND OVER HOUSING DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL HOUSING DIAMETERS AND OF BERYLLIUM COPPER.

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(f) VIBRATION = A RETAINING RING WILL BE SECURE AGAINST VIBRATION IF ITS SHEAR FATIGUE STRENGTH EQUALS OR EXCEEDS THE FORCE CREATED BY THE MASS OF THE PART BEING REPEATEDLY ACCELERATED AGAINST THE RING.

(1) SHEAR STRENGTH OF RING (CRITICAL). (ABUTTING COMPONENTS TO HAVE SHARP CORNERS).

$$XT\ 1.1\ H \geq \frac{W}{g}\ a$$

OR FOR HARMONIC MOTION

$$a = \delta\omega^2$$

$$XT\ 1.1\ H \geq \frac{W}{g}\ \delta\omega^2$$

WHERE:

X = ULTIMATE SHEAR STRENGTH OF RING MATERIAL (PSI) 1/
T = RING THICKNESS (INCHES)
H = HOUSING DIAMETER (INCHES)
W = WEIGHT OF ABUTTING PARTS (POUNDS)
g = ACCELERATION DUE TO GRAVITY (IN/SEC^2)
δ = AMPLITUDE OF VIBRATION (INCHES)
ω = ANGULAR SPEED (RAD/SEC)
a = ACCELERATION ($IN.SEC^2$)

(2) COMPRESSIVE STRENGTH OF GROOVE MATERIAL (CRITICAL) =

$$1.65YdH \geq \frac{W}{g}\ \delta\omega^2$$

WHERE:

Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL. (PSI)
d = GROOVE DEPTH (INCHES)
OTHER SYMBOLS AS SHOWN IN (1) ABOVE

(g) IMPACT CAPACITY OF RING OR GROOVE WALL =

$$I_R = \frac{PT}{2}$$ FOR THE RING (INCH POUNDS) (ABUTTING COMPONENTS TO HAVE SHARP CORNERS)

$$I_G = \frac{Pd}{2}$$ FOR THE GROOVE (INCH POUNDS)

WHERE:

P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS)
T = RING THICKNESS (INCHES)
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
d = GROOVE DEPTH (INCHES)
I_R = IMPACT CAPACITY OF RING (INCH POUNDS)

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(h) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED =

WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULA ON PAGE 6 WILL APPLY. THE CORNER RADII AND CHAMFERS LISTED ON PAGE 2 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE V.

IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH THE FOLLOWING FORMULAS:

$$P^1 = \frac{P\,CH}{CH^1}$$

OR

$$P^1 = \frac{PR}{R^1}$$

WHERE:

P^1 = NEW ALLOWABLE THRUST LOAD
P = LISTED ALLOWABLE THRUST LOAD
CH^1 = NEW (SMALLER) CHAMFER.
CH = LISTED CHAMFER.
R^1 = NEW (SMALLER) CORNER RADIUS.
R = LISTED CORNER RADIUS.

LIMIT LOADS LISTED IN TABLE V ARE BASED ON RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL (WORKING STRESS = 250,000 P.S.I.) AND OF BERYLLIUM COPPER (WORKING STRESS = 180,000 PSI). IF THE ALLOWABLE GROOVE CAPACITY LOADS, AS CALCULATED BY USING THE FORMULA ON PAGE 6 ARE LESS, THEN THEY SHOULD BE USED.

TABLE V. LIMIT LOADS.

NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS.	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.250	.312	190 LB	140 LB
.375	.453	530 LB	380 LB
.500	.750	1100 LB	800 LB
.777	.938	1650 LB	1200 LB
1.000	1.500	2400 LB	1700 LB

1/ X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .750 INCH HOUSING DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .777 INCH AND OVER HOUSING DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL HOUSING DIAMETERS AND OF BERYLLIUM COPPER.

4. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

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MILITARY INTEREST

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-002)

Review activities:
Army – AV, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
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